



Unlocking Equity & Efficiency

India's Crop Insurance Schemes Performance Review

2018 - 2022 Analysis



Executive Summary

Comprehensive analysis of **6,161 district-level entries** across **625 districts** and **26 states**, covering India's flagship crop insurance programs: PMFBY and WBCIS. The study reveals significant growth in policy adoption (+80%) alongside persistent equity challenges in gender and caste inclusion.

6,161

DISTRICT
RECORDS

+80%

POLICY
GROWTH

97.8%

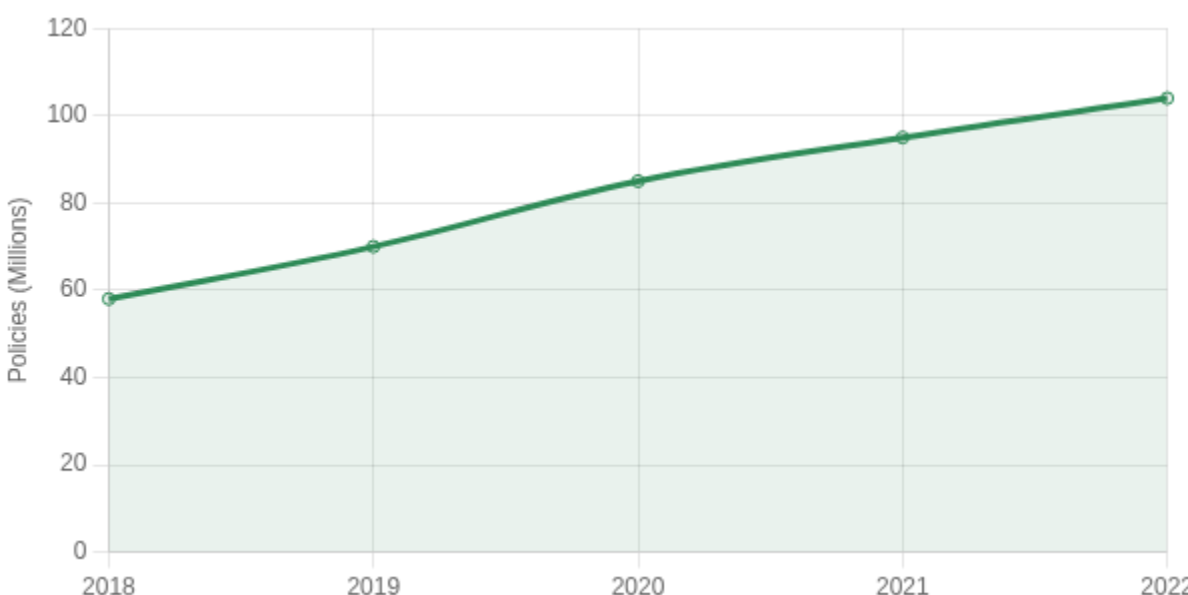
PMFBY
DOMINANCE

81.1%

SMALL+MARGINAL
FARMERS



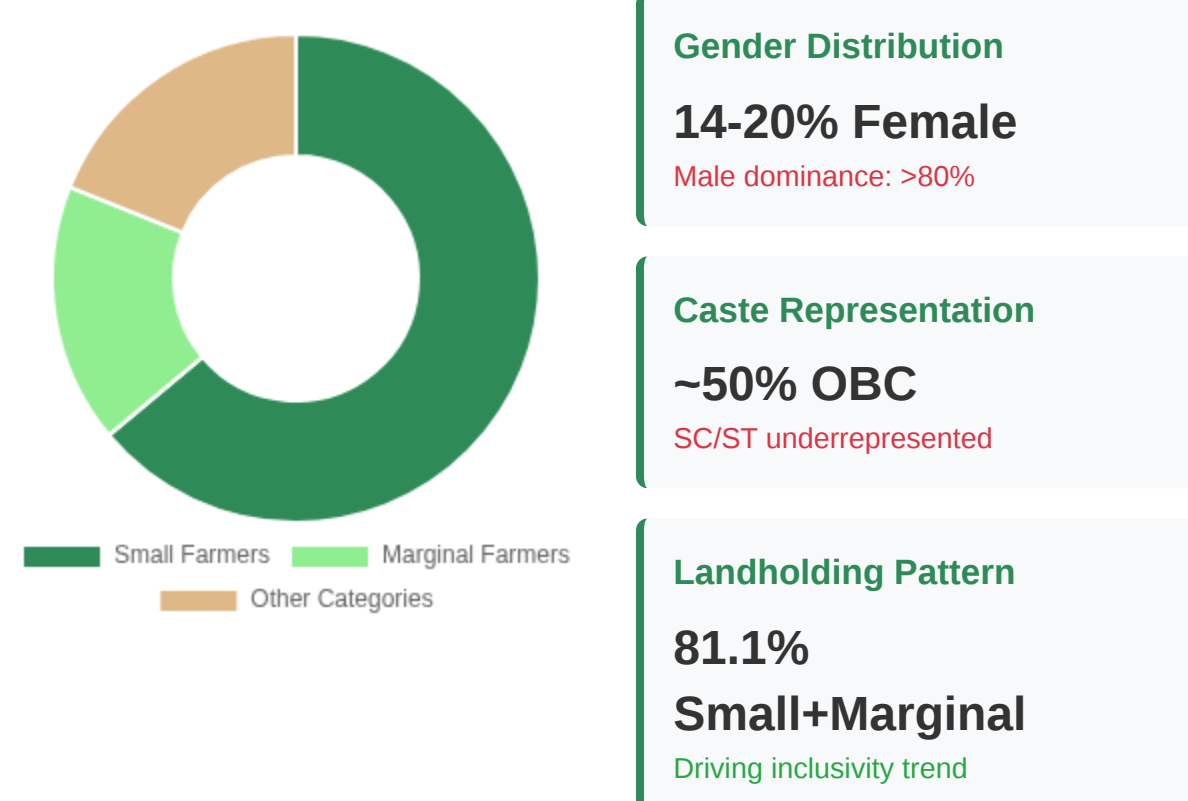
National Scale Transformation



Key Insight: The "Vulnerable Farmer Paradox" - while policy numbers surged from 58M to 104M, average land area and sum insured per policy declined as schemes penetrated smaller holdings.



Demographic Inclusivity Profile



Data Quality Note: ~1% demographics imputed, 739 district codes filled via name matching, zero-premium records require cleaning for accurate efficiency metrics.



Geographic Performance Landscape



Insurance Heartland: High-volume core belt spans Maharashtra, Rajasthan, Madhya Pradesh, and Karnataka

Volume Leaders

Rajasthan Most policies issued
Maharashtra Highest avg premium
Madhya Pradesh Consistent participation

Value Champions

Telangana Highest sum/policy
Haryana High value coverage
Karnataka Balanced growth

Inclusion Gaps

Northeast States Low participation
Himalayan Belt Geographic barriers
Gujarat (parts) Uneven coverage



District-Level Insights

Policy Hotspots (>5M policies)

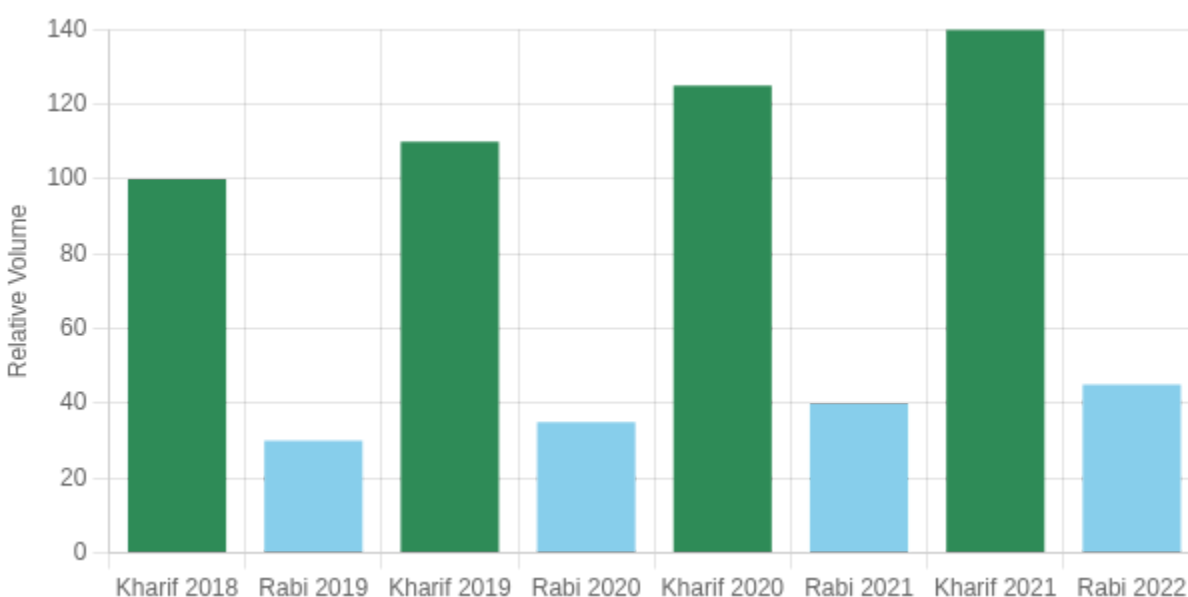
Hanumangarh Rajasthan - Agricultural hub
Beed Maharashtra - Cotton belt
Sri Ganganagar Rajasthan - Canal irrigation

High-Value Districts (>₹100K/policy)

Nuh Haryana - Premium crops
Nellore Andhra Pradesh - Rice intensive
Warangal Telangana - Commercial farming



Seasonality & Efficiency Patterns



Sawtooth Pattern

Confirmed: Kharif peaks and Rabi troughs dominate the agricultural insurance cycle, reflecting natural cropping seasons.



Correlation Insights:

Premium vs. Sum Insured shows strong correlation ($r \approx 0.88$), while demographics show negligible impact on uptake patterns.



Strategic Recommendations

- **Bridge Gender Gap:** Targeted outreach and credit access for women in Haryana, Rajasthan, UP, West Bengal
- **Boost SC/ST Participation:** Tailored subsidies in Bihar, Odisha, Jharkhand, Chhattisgarh
- **Regional Expansion:** Pilot programs in Northeast and Himalayan states
- **Data Quality Overhaul:** Clean zero-premium outliers, implement robust per-policy benchmarks
- **Landholding-Sensitive Design:** Policy variants for small/marginal-dominant states
- **Seasonal Risk Mitigation:** Multi-season triggers and weather-index adjustments

Data Source: 6,161 district-level entries | Coverage: 625 districts, 26 states | Analysis Period: 2018-2022

Generated from comprehensive analysis of PMFBY and WBCIS performance data